



ZRC-721: Zoo Non-Fungible Token Standard with Yield Transferability and Conservation Semantics

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Abstract

We present **ZRC-721**, the canonical non-fungible token standard for the Zoo L2 chain (Chain ID: 200200). ZRC-721 is fully ERC-721 compatible and extends the interface with three Zoo-specific mechanisms rooted in the project’s 2021–2022 origins: (1) **yield transferability until burn**, where NFT yield and collateral transfer to new owners upon sale and are realized (claimed) when the animal is “freed” (burned)—a pattern from the January 2022 \$ZOO deck that predates ERC-6551 token-bound accounts by over a year; (2) **generation-based rarity**, preserving the original 16 generation-one base animals from the October 2021 collection (later expanded to 20); and (3) **conservation impact attribution**, where each NFT tracks cumulative conservation contributions as non-transferable metadata. The standard also supports cross-chain bridging via Lux Warp messaging with yield state preservation, continuing the “play on any chain” philosophy established by the February 2022 ZOO Bridge.

Keywords: non-fungible token, ERC-721, yield transferability, conservation, token-bound accounts, Zoo Network

1 Introduction

Zoo NFTs predate the current Zoo L2 chain. The original collection launched on BSC in Q4 2021, integrated with PancakeSwap for \$ZOO token liquidity and egg purchases [1]. The October 2021 deck described the collection:

“Contains 1 of 16 Generation one Base Animals.”

These 16 generation-one base animals formed the initial collection, later expanded to 20 in subsequent materials. Each NFT was an ERC-721 token on BSC with a distinctive property: **yield was transferable via NFT ownership**. The January 2022 deck stated [2]:

“NFT yield and collateral is transferable to new owners until the Animal is freed from ZOO.”

Here, “freeing” an animal from Zoo corresponds to burning the NFT, which realizes all accumulated yield and collateral. This yield-transferability-until-burn pattern was a significant innovation: it meant that Zoo NFTs were not merely collectibles but yield-bearing instruments whose economic value transferred with ownership. This is precisely the semantic pattern later formalized by ERC-6551 token-bound accounts (May 2023) [4], making Zoo’s 2022 implementation a functional predecessor by more than a year.

ZRC-721 formalizes this pattern as a standard for the Zoo L2 chain, preserving backward compatibility with ERC-721 while encoding yield transferability, generation-based rarity, and conservation semantics directly in the token interface.

1.1 Historical Context

1.2 Contributions

1. **Yield Transferability:** Formalization of the 2022 yield-transferability-until-burn pattern as a standard interface (Section 2).
2. **Generation System:** Encoding the original 16 generation-one base animals and the expansion mechanism (Section 3).
3. **Conservation Hooks:** Per-NFT impact attribution tracking (Section 4).

Date	Milestone	Source
Oct 2021	16 generation-one base animals on BSC	[1]
Oct 2021	PancakeSwap integration for \$ZOO + eggs	[1]
Oct 2021	“Bridge between ETH and Binance” roadmapped	[1]
Jan 2022	“NFT yield transferable to new owners until freed”	[2]
Feb 2022	ZOO Bridge live: NFTs bridgeable across chains	[2]
May 2023	ERC-6551 proposed (Zoo pattern predates by 16 months)	[4]
2025	Migration to Zoo L2 (chain ID 200200)	This paper
2026	ZRC-721 standard formalized	This paper

Table 1: Zoo NFT chronology. The yield-transferability-until-burn pattern (January 2022) predates ERC-6551 by 16 months.

4. **Bridge Compatibility:** Yield-preserving cross-chain transfers via Lux Warp (Section 5).
5. **Formal Specification:** Complete Solidity interface and invariant analysis (Section 6).

2 Yield Transferability Until Burn

2.1 The 2022 Pattern

The original Zoo NFT mechanism on BSC implemented the following:

1. Each NFT N accrues yield $Y_N(t)$ from Zoo staking pools over time t .
2. On transfer of N from owner A to owner B , the accumulated yield $Y_N(t)$ transfers with the NFT. Owner B inherits all rights to the accumulated yield.
3. Collateral locked against N also transfers with ownership.
4. When the animal is “freed” (burned), the owner at burn time receives all accumulated yield $Y_N(t)$ and all collateral.
5. A freed animal cannot be transferred, bridged, or yield further.

This is precisely a token-bound account: the NFT “owns” yield and collateral, which transfer with the token.

2.2 ZRC-721 Yield Interface

```

1 interface IZRC721Yield {
2     /// @notice Returns accumulated yield for token
3     /// @param tokenId The NFT to query
4     /// @return amount Accumulated yield in ZOO wei
5     function accumulatedYield(uint256 tokenId)
6         external view returns (uint256 amount);
7
8     /// @notice Returns collateral locked against token
9     /// @param tokenId The NFT to query
10    /// @return amount Locked collateral in ZOO wei
11    function lockedCollateral(uint256 tokenId)

```

```

12         external view returns (uint256 amount);
13
14     /// @notice Free the animal: burn NFT, realize yield + collateral
15     /// @param tokenId The NFT to free
16     /// @dev Transfers accumulatedYield + lockedCollateral to msg.sender
17     /// @dev Burns the NFT permanently
18     function free(uint256 tokenId) external;
19
20     /// @notice Check if animal has been freed
21     /// @param tokenId The NFT to query
22     /// @return True if freed (burned with yield realized)
23     function isFreed(uint256 tokenId)
24         external view returns (bool);
25
26     /// @notice Emitted when yield is accumulated
27     event YieldAccumulated(
28         uint256 indexed tokenId,
29         uint256 amount,
30         uint256 totalAccumulated
31     );
32
33     /// @notice Emitted when animal is freed
34     event AnimalFreed(
35         uint256 indexed tokenId,
36         address indexed owner,
37         uint256 yieldRealized,
38         uint256 collateralReleased
39     );
40 }

```

Listing 1: ZRC-721 Yield Interface

2.3 Yield Transfer on Sale

When an ERC-721 `transferFrom` or `safeTransferFrom` is executed, the yield state is *not reset*. The new owner inherits the full accumulated yield:

Definition 2.1 (Yield Transfer Invariant). *For any ZRC-721 token N with accumulated yield Y_N and collateral C_N , a transfer from owner A to owner B preserves:*

$$Y_N^{\text{after}} = Y_N^{\text{before}} \quad (1)$$

$$C_N^{\text{after}} = C_N^{\text{before}} \quad (2)$$

The new owner B has exclusive rights to Y_N and C_N .

Theorem 2.1 (Yield Conservation). *For any sequence of transfers and yield accumulations, the yield realized on burn equals the total yield accumulated over the NFT’s lifetime:*

$$Y_{\text{realized}} = \sum_{i=0}^T \Delta Y_i$$

where ΔY_i is yield accumulated in epoch i and T is the burn epoch.

Proof. Yield is monotonically non-decreasing (only accumulation, no withdrawal before burn) and transfer-invariant (no yield is lost or gained on transfer). The burn operation (“free”) is the only yield realization event. Therefore, realized yield equals cumulative accumulation. $\square$

Generation	Base Animals	Source
Gen 1 (Oct 2021)	16	Original collection
Gen 1 expanded	20	Later materials
Gen 2+	Variable	Governance-approved

Table 2: Zoo NFT generations. The original 16 base animals form the canonical Gen 1 collection.

3 Generation System

3.1 Original 16 Base Animals

The October 2021 collection comprised 16 generation-one base animals:

```

1 interface IZRC721Generation {
2     /// @notice Returns the generation of the token
3     /// @param tokenId The NFT to query
4     /// @return gen Generation number (1 = original base animals)
5     function generation(uint256 tokenId)
6         external view returns (uint256 gen);
7
8     /// @notice Returns the species identifier
9     /// @param tokenId The NFT to query
10    /// @return species Species string (e.g., "base_animal_01")
11    function species(uint256 tokenId)
12        external view returns (string memory species);
13
14    /// @notice Returns total species count for a generation
15    /// @param gen Generation number
16    /// @return count Number of species in generation
17    function speciesCount(uint256 gen)
18        external view returns (uint256 count);
19 }

```

Listing 2: ZRC-721 Generation Interface

3.2 Egg Mechanism

The original BSC deployment used an egg mechanism integrated with PancakeSwap:

1. User purchases eggs using \$ZOO via PancakeSwap
2. Each egg hatches into one of the 16 generation-one base animals
3. Hatching is random with equal probability across species
4. The hatched NFT immediately begins accruing yield

ZRC-721 preserves the egg mechanism as an optional minter extension.

4 Conservation Impact Attribution

Each ZRC-721 NFT tracks cumulative conservation contributions:

```

1 interface IZRC721Conservation {
2     /// @notice Returns cumulative conservation impact
3     /// @param tokenId The NFT to query
4     /// @return amount Total conservation contribution in ZOO wei
5     function conservationImpact(uint256 tokenId)
6         external view returns (uint256 amount);
7
8     /// @notice Record a conservation contribution
9     /// @param tokenId The NFT crediting the contribution
10    /// @param amount Contribution amount in ZOO wei
11    /// @dev Only callable by verified conservation fund
12    function recordImpact(uint256 tokenId, uint256 amount)
13        external;
14
15    /// @notice Emitted on conservation contribution
16    event ConservationContribution(
17        uint256 indexed tokenId,
18        uint256 amount,
19        uint256 totalImpact
20    );
21 }

```

Listing 3: ZRC-721 Conservation Interface

Conservation impact is **non-transferable**: it remains bound to the NFT (not the owner) and survives transfers. This creates a provenance record of the NFT’s conservation history.

5 Cross-Chain Bridge Compatibility

5.1 Historical: ZOO Bridge (2022)

The February 2022 ZOO Bridge [2] was one of the earliest cross-chain NFT bridges, described as allowing “each users to play ZOO on any chain.” It listed 16 supported asset types spanning EVM and non-EVM chains. ZRC-721 is designed for bridge compatibility from the start, continuing this tradition.

5.2 Warp Bridge Interface

ZRC-721 tokens are bridgeable via Lux Warp messaging with yield state preservation:

```

1 interface IZRC721Bridge {
2     /// @notice Encode token for bridge transfer
3     /// @param tokenId The NFT to bridge
4     /// @return payload Warp-compatible payload including yield state
5     function encodeBridgePayload(uint256 tokenId)
6         external view returns (bytes memory payload);
7
8     /// @notice Decode and mint bridged token
9     /// @param payload Warp message payload
10    /// @param recipient Destination owner
11    /// @return tokenId Minted wrapped token ID
12    function decodeBridgeMint(
13        bytes calldata payload,
14        address recipient
15    ) external returns (uint256 tokenId);
16 }

```

```

17     /// @notice Emitted on bridge lock
18     event BridgeLocked(
19         uint256 indexed tokenId,
20         uint256 destChainId,
21         bytes payload
22     );
23
24     /// @notice Emitted on bridge mint
25     event BridgeMinted(
26         uint256 indexed tokenId,
27         uint256 srcChainId,
28         address indexed recipient
29     );
30 }

```

Listing 4: ZRC-721 Bridge Interface

The bridge payload includes the full yield state, generation, species, and conservation impact, ensuring that bridged NFTs retain all ZRC-721 semantics on the destination chain.

5.3 Yield Preservation Across Chains

Definition 5.1 (Bridge Yield Invariant). *For a ZRC-721 token N bridged from chain A to chain B :*

1. N is locked in escrow on chain A ; wrapped N' minted on chain B
2. $Y_{N'} = Y_N$ at time of bridge (yield state preserved)
3. N' continues accruing yield on chain B
4. Freeing N' on chain B (burn) triggers yield realization and releases N on chain A
5. Total yield: $Y_N + \Delta Y_{N'}$ (no yield lost)

This directly implements the 2022 deck’s guarantee: yield is transferable (including across chains) until the animal is freed.

6 Formal Specification

6.1 Complete Interface

```

1  interface IZRC721 is
2      IERC721,
3      IERC721Metadata,
4      IERC721Enumerable,
5      IZRC721Yield,
6      IZRC721Generation,
7      IZRC721Conservation,
8      IZRC721Bridge
9  {
10     /// @notice Returns the ZRC-721 standard version
11     function zrcVersion() external pure returns (string memory);
12 }

```

Listing 5: ZRC-721 Complete Interface

Property	Zoo NFTs (Jan 2022)	ERC-6551 (May 2023)
NFT owns assets	Yield + collateral	Arbitrary tokens
Transfer semantics	Assets move with NFT	Account moves with NFT
Realization	Burn (“free the animal”)	Direct withdrawal
Composability	Yield-specific	General-purpose
Timeline	January 2022	May 2023

Table 3: Zoo’s yield-transferability pattern vs. ERC-6551. Zoo’s implementation predates ERC-6551 by 16 months.

6.2 Invariants

1. **Yield monotonicity:** $Y_N(t + 1) \geq Y_N(t)$ for all tokens N and times $t < t_{\text{burn}}$.
2. **Transfer preservation:** Y_N and C_N are unchanged by `transferFrom`.
3. **Burn finality:** After `free(tokenId)`, the token cannot be transferred, bridged, or yield further.
4. **Conservation permanence:** Conservation impact is non-transferable and monotonically non-decreasing.
5. **Generation immutability:** A token’s generation and species cannot be changed after minting.

7 Relation to ERC-6551

ERC-6551 [4] (proposed May 2023) creates token-bound accounts: smart contract accounts owned by NFTs. Zoo’s yield-transferability-until-burn pattern (January 2022) achieves equivalent semantics through a different mechanism:

ZRC-721 acknowledges this lineage by making yield transferability a first-class interface rather than an add-on extension.

8 Related Work

Zoo Oct 2021 Deck: Original collection of 16 generation-one base animals on BSC with PancakeSwap integration [1]. The Q4 roadmap included “Bridge between ETH and Binance.”

Zoo Jan 2022 Deck: Defined yield transferability: “NFT yield and collateral is transferable to new owners until the Animal is freed from ZOO” [2]. Confirmed February 2022 ZOO Bridge launch.

ERC-721: The base non-fungible token standard [3]. ZRC-721 is fully compatible.

ERC-6551: Token-bound accounts [4]. Zoo’s yield-transferability-until-burn pattern is a functional predecessor.

ERC-4907: Rental NFTs with time-limited usage rights. ZRC-721’s yield mechanism is orthogonal: yield accrues to the owner (not a renter), and realization requires burn.

9 Conclusion

ZRC-721 formalizes the non-fungible token standard for Zoo L2, encoding three mechanisms with roots in Zoo’s 2021–2022 origins: yield transferability until burn (from the January 2022 \$ZOO deck, predating ERC-6551 by 16 months), generation-based rarity (preserving the original 16 generation-one base animals from October 2021), and conservation impact attribution. The standard is fully ERC-721 compatible, bridge-ready via Lux Warp messaging (continuing the “play on any chain” philosophy of the February 2022 ZOO Bridge), and designed to preserve the yield-transferability-until-burn pattern that made Zoo NFTs yield-bearing instruments before the concept was standardized.

Acknowledgments

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References

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